

Muhammad Abdullah

Date of birth: 13/11/2005 | **Nationality:** Pakistani | **Phone number:** (+92) 3094157102 (Mobile) | **Email address:** abdullahse2211@gmail.com | **LinkedIn:** Muhammadabdullahse2211 | **GitHub:** abdullahse2211 | **My_Portfolio :** Muhammad Abdullah | **Address:** 53400, Lahore, Pakistan (Home)

EDUCATION AND TRAINING

02/11/2022 – 10/08/2026 Lahore, Pakistan
BACHELOR OF SOFTWARE ENGINEERING Lahore Garrison University, DHA, Lahore

Machine Learning, Probability & Statistics, Stochastic Processes, Linear Algebra, Calculus, Data Structures & Algorithms, Big Data Analytics, Operational Research.

Final grade 3.0/4 | **Level in EQF** EQF level 6 |

Thesis RetinaXplain: A Hybrid CNN–Vision Transformer Framework for Diabetic Retinopathy Classification

WORK EXPERIENCE

AI/ML ENGINEER (FULL-TIME) – CODIUM HUB – 01/09/2025 – Current – LAHORE, PAKISTAN

Medical & Edge Computer Vision: Engineered and benchmarked hybrid CNN-Transformer architectures (Swin-UNet, MedSAM) and PyTorch backbones for 2D/3D medical image segmentation and real-time edge tracking (OpenCV, TensorRT), achieving sub-second GPU latency and 30+ FPS on embedded nodes.
High-Performance MLOps & Inference Optimization: Architected production-grade MLOps infrastructure leveraging vLLM and TensorRT-LLM with AWQ/QAT quantization, reducing VRAM consumption by 45% while doubling microservice inference throughput.
Enterprise Knowledge Retrieval & GraphRAG: Developed advanced Graph Neural Network (GNN) integrated GraphRAG pipelines and automated data preprocessing workflows, boosting complex structured dataset retrieval accuracy by 35%.

AI/ML ENGINEER (INTERN) – CODIUM HUB – 01/06/2025 – 31/08/2025 – LAHORE

Data Engineering & Preprocessing: Built automated data preprocessing and synthetic augmentation pipelines in PyTorch for deep learning models.
Baseline Architecture & Scripting: Developed initial benchmarking scripts for U-Net, ResNet, and EfficientNet architectures.
Evaluation Metrics & Workflows: Established baseline evaluation metrics for downstream computer vision tasks and assisted with dataset engineering workflows.

MACHINE LEARNING & DEEP LEARNING (INTERN) – ARTIFICIAL INTELLIGENCE TECHNOLOGY CENTER (AITEC), (NCP) – 27/05/2024 – 27/08/2024 – ISLAMABAD, PAKISTAN

Core Machine Learning: Developed Python and PyTorch-based ML/DL models for experimental data analysis and pattern recognition.
Neural Network Architecture: Built and tuned foundational neural networks (ANNs/CNNs) for dataset processing and feature extraction.
Research Benchmarking: Conducted AI research tasks at AITEC (NCP), optimizing model parameters and documenting evaluation metrics.

PUBLICATIONS

2026
RetinaXplain: An Explainable Multi-Model Deep Learning Framework for Multi-Class Diabetic Retinopathy Severity Grading

Role: Lead Author & Primary Investigator. Developed an end-to-end multi-model hybrid deep learning pipeline utilizing Swin-T for diabetic retinopathy grading, achieving 85.27% accuracy. Integrated automated diagnostic report generation and multi-model inference, documented in a 72-page technical report.

Manuscript in Preparation.

Authors: Muhammad Abdullah, Omer Aftab | **Journal Name**: MDPI Bioengineering | **Publisher**: MDPI

2026
Fair Contestability Budgeting under Audit Uncertainty: Allocating Scarce Review on Real Credit-Risk Data

Status: Under peer review.

Contributions: Review of Literature, proof read and edit, Visualization analysis.

Authors: Hamza Shahbaz, Muhammad Abdullah | **Journal Name**: Advances in Artificial Intelligence Research (DergiPark) | **Publisher**: DergiPark

ACADEMIC & AI LAB RESEARCH PROJECTS

01/01/2026 – 22/05/2026
RetinaXplain - Hybrid CNN-Vision Transformer Framework for Diabetic Retinopathy Classification

Final Year Project-*Sole AI Researcher and Developer*
Data Pipeline & Preprocessing: Processed 3,662 APTOS images across 5 DR classes using circular cropping and Ben Graham normalization, expanding data to ~16,000 augmented instances.
Model Training & Algorithmic Features: Trained and evaluated Swin-T (85.27% accuracy, 0.7216 macro-F1), EfficientNet-B3, and ViT-B/16, engineering weighted soft-voting ensembles, clinical label remap,& agreement analysis.
App Deployment & Documentation: Built a multi-page web app featuring real-time DR diagnostic reports, probability visualizers, session analytics, and downloadable JSON records; authored a 72-page technical report.

Links https://github.com/abdullahse2211/diabetic-retinopathy-classification | https://youtu.be/MnS1vS5ZLqU?si=td3M1nW6vMmZRVf5

07/01/2026 – 08/03/2026
BioNAFNet - Low-Light Image Enhancement | Student Deep Learning Researcher

Novel Architecture & Loss Formulation: Designed a bio-inspired enhancement network trained from scratch with a composite Charbonnier, SSIM, color, and Weber-Fechner perceptual loss, surpassing 9 baselines with 28.67 dB PSNR on LOLv2 Real.
Knowledge Distillation & Acceleration: Compressed model parameters from 154M to 24M via knowledge distillation (6.4x compression, 3.9x speedup) with zero perceptual degradation, enabling the student model to surpass the teacher by +0.29 dB.

01/12/2025 – 02/02/2026
ThalCare - Alpha Thalassemia Classification | Student ML Researcher

Clinical Feature Engineering: Engineered 15 diagnostic features combining 9 CBC/HPLC parameters with 6 clinical indices (Mentzer, Shine & Lal, Green & King), elevating Silent Carrier diagnostic F1 to 0.92–0.955.
Robust Multi-Class ML Benchmarking: Evaluated 6 ML architectures (hybrid CNN-Transformer, 1D-CNN, SVM, RF, LR) for 5-class differential diagnosis, achieving 90.60% balanced accuracy and 0.906 weighted F1 under injected clinical noise and 10% feature dropout.

Links https://github.com/abdullahse2211/ThalCare.git | https://youtu.be/ciiGgdlZ4d4?si=_xKxSkJOZ9t3njEA

LEADERSHIP, HACKATHONS & MENTORSHIP

04/03/2026 – 05/03/2026
Flagship AI Hackathon Director

Spearheaded and personally directed the department's flagship AI Hackathon, orchestrating technical infrastructure, real-time submission evaluation pipelines, and jury coordination for 50+ participants across 15 teams.

04/01/2026 – 31/05/2026
Peer Mentoring & Technical Support

Actively mentored junior lab members on advanced PyTorch workflows, GPU optimization techniques, and Git-based collaborative development.

02/09/2025 – 08/06/2026
Departmental Event Organization

Assisted in organizing technical guest lectures featuring software industry executives and supported student exposure visits to local tech firms.

SKILLS

AI, Vision & Frameworks:

PyTorch, TensorFlow, OpenCV, scikit-learn, MedSAM, Swin-Transformer, UNet3+, Diffusion, RAG

Systems, MLOps & GenAI:

VLLM, TensorRT, Triton, Docker, 4-bit AWQ, Qdrant, GraphRAG, Qwen2-VL, LlamaIndex, LoRA, BoTorch

Languages & Tools:

Python, C++, SQL (MySQL), MATLAB, NumPy, Pandas, Matplotlib, Streamlit, Git, GitHub, Overleaf

CERTIFICATIONS

University of Engineering and Technology, Lahore National Centre of Artificial Intelligence, 08/12/2023
Prompt Engineering - Certificate of Completion

University of Management and Science (LUMS), 22/12/2024
INTRODUCTION TO DATA SCIENCE

PERSON VUE CERTIPORT, 13/01/2025
Information Technology Specialist: JavaScript

International industry certification in JavaScript. Credential ID: mTp4-DwVM

Link https://verify.certiport.com/

National Vocational & Technical Training Commission (NAVTTC), 16/12/2024
JavaScript Full-Stack Development (MERN)

IBM; Coursera, 18/11/2023
IBM Web Development

Link https://coursera.org/verify/E8X42H6HASJL

CONFERENCES AND SEMINARS

24/07/2023 – 25/07/2023 Lahore
International Symposium on Artificial Intelligence Medical Imaging and Pattern Recognition

12/08/2025 – 14/08/2025 Lahore
Computer Vision & Edge Intelligence Summit

17/06/2025 – 18/06/2025
Advanced MLOps & Large Language Models Workshop

16/11/2025 – 16/11/2025 Lahore
Data Science & GraphRAG Technical Seminar

02/06/2026 – 02/06/2026 Lahore
Next-Gen AI Research & Vision Systems Colloquium

06/11/2023 – 06/11/2023 Lahore
Scientific Writing for Impact Factor Journals 1.0 CPD Credit Point

13/11/2023 – 13/11/2023 Lahore
Tools to Improve Your Professional Writing 1.0 CPD Credit Point

15/10/2024 – 15/10/2024 Lahore
Information Technology Career Symposium 2024

LANGUAGE SKILLS

Mother tongue(s): **URDU**

Other language(s): **ENGLISH**